

NovaPharma Iberia — Step 3.A Review: Python Pipeline Prototype

Digital transformation project · Excel → MySQL / Python / Power BI / Anaplan · Author: Nix · 20 September 2026

What was done

A Jupyter notebook that takes the data from the Excel model, cleans it and loads it into the MySQL warehouse created in Step 2. It is a prototype: it explores the data and takes decisions once, against the historical file. The monthly production script (Step 3.B) will reuse the same cleaning rules without the exploration. Six stages, one after the other:

Stage	What it does	Result
1 Extract	Read the 8 data sheets of the Excel into Python tables (DataFrames)	8 tables, row counts match the inventory
2 Profile	Look at Sales_Raw without changing anything: types, empty values, ranges, unique IDs, codes that exist (or not) in the masters	6 findings (table below)
3 Rules	Decide what to do with each finding. No code: a decision table	8 rules — fix, quarantine or keep
4 Transform	Apply the rules; give every table the shape of the SQL model (names, columns, new dim_brand, Budget + Forecast merged into fact_plan)	48,100 clean sales rows, 63 quarantined
5 Load	Write the tables into MySQL, dimensions first, then facts, then the quarantine	9 tables loaded, no rejections
6 Reconcile	Check that SQL totals equal Excel totals minus the quarantine; re-run the 11 Excel quality checks in SQL	Match to the cent; 8 OK, 3 INFO, 0 fails

What the profiling found

#	Finding	Rows	Decision
1	Negative units and sales (returns / credit notes)	60	Keep — valid business case
2	Duplicate transaction IDs	45	Quarantine
3	SKU codes with a space at the end	25	Fix (strip)
4	SKU codes in lower case — Excel did not see them because VLOOKUP ignores case	70	Fix (upper-case)
5	Customer code C9999 does not exist in the customer master	18	Quarantine
6	Dates: 32 months, Jan-2024 to Aug-2026, none in the future	—	Keep

Three ideas to remember

Where the cleaning rules come from. First from the data model (every primary key, foreign key and NOT NULL in the DDL is a rule); second from profiling (what fails today); third from the business (what only a person can decide, e.g. returns are valid). The Excel sheet Data_Quality was the business knowledge already written down, and was used as the checklist.

Nothing is deleted. Rows that cannot enter the warehouse are set aside in a quarantine table with the reason. If the business later confirms C9999 as a real customer, the 18 rows can be reloaded. Deleting is the only operation that cannot be undone.

Reconcile, do not assume. Excel net sales 429.1 M€ minus quarantine 0.84 M€ = SQL net sales 428.3 M€, exactly. And the same 11 quality checks that read "FAIL: 5" in Excel now read OK in SQL, on the same data.

Vocabulary

ETL = Extract, Transform, Load (the L is Load; cleaning is part of Transform). Reconciliation = conciliación (compare two sources and explain every difference); consolidation = consolidación (add several sources into one view). Profiling reads, transforming writes. A notebook is for designing; a script is for running every month.